

Coordinate Geometry (10 th std)

Choose the Appropriate Answer

1. The area of triangle formed by the points $(-5, 0)$, $(0, -5)$ and $(5, 0)$ is
 (a) 0 sq. Units (b) 25 sq. Units (c) 5 sq. Units (d) None of these

Ans: (b) 25 sq. Units

$$\begin{aligned} \text{Area of the } \Delta &= \frac{1}{2} \begin{vmatrix} -5 & 0 & 5 \\ 0 & -5 & 0 \\ 0 & 0 & 0 \end{vmatrix} \\ &= \frac{1}{2} [25 + 0 + 0 - (0 - 25 + 0)] = 25 \text{ sq. units.} \end{aligned}$$

2. A man walks near a wall, such that the distance between him and the wall is 10 units. Consider the wall to be the Y axis. The path travelled by the man is
 (a) $x = 10$ (b) $y = 10$ (c) $x = 0$ (d) $y = 0$

Ans: (a) $x = 10$

3. The straight line given by the equation $x = 11$ is
 (a) Parallel to X axis (b) Parallel to Y axis
 (c) Passing through the origin (d) Passing through the point $(0, 1)$

Ans: (b) Parallel to Y axis

4. If $(5, 7)$, $(3, p)$ and $(6, 6)$ are collinear, then the value of p is
 (a) 3 (b) 6 (c) 9 (d) 12

Ans: (c) 9

Since the three points are collinear. Area of a triangle is 0

$$\frac{1}{2} \begin{vmatrix} 5 & 3 & 6 \\ 7 & p & 6 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

$$5p + 18 + 42 - (21 + 6p + 30) = 0 \Rightarrow p = 9$$

5. The point of intersection of $3x - y = 4$ and $x + 7 = 8$ is
 (a) $(5, 3)$ (b) $(2, 4)$ (c) $(3, 5)$ (d) $(4, 4)$

Ans: (c) $(3, 5)$

$$\begin{aligned} 3x - y &= 4 \quad \dots(1) \\ x + y &= 8 \quad \dots(2) \end{aligned}$$

$$(1) + (2) \Rightarrow 4x = 12 \Rightarrow x = \frac{12}{4} = 3$$

Substitute the value of $x = 3$ in (2)
 $3 + 7 = 8 \Rightarrow y = 8 - 3 = 5$
 The point of intersection is $(3, 5)$

6. The slope of the line joining $(12, 3)$, $(4, a)$ is 18. The value of 'a' is
 (a) 1 (b) 4 (c) -5 (d) 2

Ans: (d) 2

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{1}{8} \Rightarrow \frac{a-3}{4-12} = \frac{1}{8} \Rightarrow \frac{a-3}{-8} = \frac{1}{8}$$

$$a-3 = -1 \Rightarrow \text{The value of } a = 2$$

7. The slope of the line which is perpendicular to a line joining the points (0, 0) and (-8, 8) is

- (a) -1 (b) 1 (c) 13 (d) -8

Ans: (b) 1

8. If slope of the line PQ is $1/\sqrt{3}$ then slope of the perpendicular bisector of PQ is

- (a) $\sqrt{3}$ (b) $-\sqrt{3}$ (c) $1/\sqrt{3}$ (d) 0

Ans: (b) $-\sqrt{3}$

9. If A is a point on the Y axis whose ordinate is 8 and B is a point on the X axis whose abscissae is 5 then the equation of the line AB is

- (a) $8x + 5y = 40$ (b) $8x - 5y = 40$ (c) $x = 8$ (d) $y = 5$

Ans: (a) $8x + 5y = 40$

10. The equation of a line passing through the origin and perpendicular to the line $1x - 3y + 4 = 0$ is

- (a) $7x - 3y + 4 = 0$ (b) $3x - 7y + 4 = 0$ (c) $3x + 7y = 0$ (d) $7x - 3y = 0$

Ans: (c) $3x + 7y = 0$

11. Consider four straight lines

- (i) $l_1 : 3y = 4x + 5$ (ii) $l_2 : 4y = 3x - 1$
 (iii) $l_3 : 4y + 3x = 7$ (iv) $l_4 : 4x + 3y = 2$

Which of the following statement is true?

- (a) l_1 and l_2 are perpendicular (b) l_2 and l_4 are parallel
 (c) l_2 and l_4 are perpendicular (d) l_2 and l_3 are parallel

Ans: (c) l_2 and l_4 are perpendicular

12. A straight line has equation $87 = 4x + 2y$. Which of the following is true

- (a) The slope is 0.5 and the y intercept is 2.6
 (b) The slope is 5 and the y intercept is 2.6
 (c) The slope is 0.5 and they intercept is 2.6
 (d) The slope is 5 and the y intercept is 2.6

Ans: (a) The slope is 0.5 and they intercept is 2.6

13. When proving that a quadrilateral is a trapezium, it is necessary to show

- (a) Two sides are parallel. (b) Two parallel and two non-parallel sides.
 (c) Opposite sides are parallel. (d) All sides are of equal length.

Ans: (b) Two parallel and two non-parallel sides.

14. When proving that a quadrilateral is a parallelogram by using slopes you must find

- (a) The slopes of two sides (b) The slopes of two pair of opposite sides
(c) The lengths of all sides (d) Both the lengths and slopes of two sides

Ans: (b) The slopes of two pair of opposite sides

15. (2, 1) is the point of intersection of two lines.

- (a) $x - y - 3 = 0$; $3x - y - 7 = 0$ (b) $x + y = 3$; $3x + y = 7$
(c) $3x + y = 3$; $x + y = 7$ (d) $x + 3y - 3 = 0$; $x - y - 7 = 0$

Ans: (b) $x + y = 3$; $3x + y = 7$

16. The points (-5, 1), (1, p) and (4, -2) are collinear if the value of p is

- (a) 3 (b) 2 (c) 1 (d) -1

Ans: (d) -1

17. The area of the triangle ABC with the vertices A(-5, 7), B(-4, -5) and C(4, 5) is

- (a) 63 (b) 35 (c) 53 (d) 36

Ans: (c) 53

18. The area of the triangle whose vertices are A(1, 2), B(-2, 3) and C(-3, -4) is

- (a) 11 (b) 22 (c) 33 (d) 21

Ans: (a) 11

19. The line segment joining the points (3, -1) and (-6, 5) is trisected. The coordinates of point of trisection are

- (a) (3, 3) (b) (-3, 3) (c) (3, -3) (d) (-3, -3)

Ans: (b) (-3, 3)

20. The line $3x + y - 9 = 0$ divides the line joining the points (1, 3) and (2, 7) internally in the ratio

- (a) 3 : 4 (b) 3 : 2 (c) 2 : 3 (d) 4 : 3

Ans: (a) 3 : 4

21. If (a / 3, 4) is the mid-point of the segment joining the points P(-6, 5) and R(-2, 3), then the value of 'a' is

- (a) 12 (b) -6 (c) -12 (d) -4

Ans: (c) -12

22. The points (1, 1), (-2, 7) and (3, -3) are

- (a) Vertices of an equilateral triangle (b) Collinear
(c) Vertices of an isosceles triangle (d) None of these

Ans: (b) Collinear

23. Two vertices of a triangle are (3, -5) and (-7, 4). If its centroid is (2, -1), then the third vertex is

- (a) (10, 2) (b) (-10, 2) (c) (10, -2) (d) (-10, -2)

Ans: (c) (10, -2)

24. If the points $P(1, 2)$, $B(0, 0)$ and $C(a, b)$ are collinear, then
(a) $2a = b$ (b) $a = -b$ (c) $a = 2b$ (d) $a = b$

Ans: (a) $2a = b$

25. The coordinates of the centroid of a triangle whose vertices are $(0, 6)$, $(8, 12)$ and $(8, 0)$ is
(a) $(4, 6)$ (b) $(16, 6)$ (c) $(8, 6)$ (d) $(16/3, 6)$

Ans: (d) $(16/3, 6)$

26. If the distance between the points $(x, -1)$ and $(3, 2)$ is 5, then the value of x is
(a) -7 or -1 (b) -7 or 1 (c) 7 or 1 (d) 7 or -1

Ans: (d) 7 or -1

27. The area of the triangle formed by the points $A(-5, 3)$, $B(6, -2)$ and $C(-3, 4)$ is
(a) 0 (b) 1 (c) 2 (d) $3/2$

Ans: (a) 0

28. The distance of the point $P(2, 3)$ from the x -axis is
(a) 2 (b) 3 (c) 1 (d) 5

Ans: (b) 3

29. The distance between the points $A(0, 6)$ and $B(0, -2)$ is
(a) 6 (b) 8 (c) 4 (d) 2

Ans: (b) 8

30. The distance of the point $P(-6, 8)$ from the origin is
(a) 8 (b) $2\sqrt{7}$ (c) 10 (d) 6

Ans: (c) 10

31. The distance between the points $(0, 6)$ and $(-6, 0)$ is
(a) 6 (b) $6\sqrt{2}$ (c) $2\sqrt{6}$ (d) 10

Ans: (b) $6\sqrt{2}$

32. The perimeter of a triangle with vertices $(0, 4)$, $(0, 0)$ and $(3, 0)$ is
(a) 5 (b) 12 (c) 11 (d) $7 + \sqrt{5}$

Ans: (b) 12

33. The points $(-4, 0)$, $(4, 0)$, $(0, 3)$ are the vertices of a
(a) Right triangle (b) Isosceles triangle
(c) Equilateral triangle (d) Scalene triangle

Ans: (b) Isosceles triangle

34. The point which divides the line segment joining the points $(7, -6)$ and $(3, 4)$ in ratio 1: 2 internally lies in the
(a) I quadrant (b) II quadrant (c) III quadrant (d) IV quadrant

Ans: (d) IV quadrant

35. The point which lies on the perpendicular bisector of the line segment joining the points A(-2, -5) and B(2, 5) is

- (a) (0, 0) (b) (0, 2) (c) (2, 0) (d) (-2, 0)

Ans: (a) (0, 0)

36. The fourth vertex D of a parallelogram ABCD whose three vertices are A(-2, 3), B(6, 7) and C(8, 3) is

- (a) (0, 1) (b) (0, -1) (c) (-1, 0) (d) (1, 0)

Ans: (b) (0, -1)

37. The points (-1, -2), (1, 0), (-1, 2), (-3, 0) form a quadrilateral of type:

- (a) Square (b) Rectangle (c) Parallelogram (d) Rhombus

Ans: (a) Square

38. If the distance between the points A(2, -2) and B(-1, x) is equal to 5, then the value of x is:

- (a) 2 (b) -2 (c) 1 (d) -1

Ans: (a) 2

39. The midpoint of a line segment joining two points A(2, 4) and B(-2, -4) is

- (a) (-2, 4) (b) (2, -4) (c) (0, 0) (d) (-2, -4)

Ans: (c) (0, 0)

40. The distance of point A(2, 4) from the x-axis is

- (a) 2 units (b) 4 units (c) -2 units (d) -4 units

Ans: (b) 4 units

41. The distance between the points P(0, 2) and Q(6, 0) is

- (a) $4\sqrt{10}$ (b) $2\sqrt{10}$ (c) $\sqrt{10}$ (d) 20

Ans: (b) $2\sqrt{10}$

42. If O(p / 3, 4) is the midpoint of the line segment joining the points P(- 6, 5) and Q(- 2, 3), the value of p is:

- (a) 7 / 2 (b) -12 (c) 4 (d) -4

Ans: (b) -12

$$P / 3 = (-6 - 2) / 2 \Rightarrow P / 3 = -4 \Rightarrow P = -12$$

43. The point which divides the line segment of points P(-1, 7) and (4, -3) in the ratio of 2:3 is:

- (a) (-1, 3) (b) (-1, -3) (c) (1, -3) (d) (1, 3)

Ans: (d) (1, 3)

By section formula we know:

$$x = [(2 \times 4) + (3 \times (-1))] / (2 + 3) = (8 - 3) / 5 = 1$$

$$y = [(2 \times (-3)) + (3 \times 7)] / (2 + 3) = (-6 + 21) / 5 = 3$$

Hence, the required point is (1, 3).

44. The ratio in which the line segment joining the points P(-3, 10) and Q(6, -8) is divided by O(-1, 6) is:

- (a) 1: 3 (b) 3: 4 (c) 2: 7 (d) 2: 5

Ans: (c) 2: 7

By the section formula, we have;
 $-1 = (6k - 3)/(k + 1) \Rightarrow -k - 1 = 6k - 3 \Rightarrow k = 2/7$

45. The coordinates of a point P, where PQ is the diameter of a circle whose centre is (2, -3) and Q is (1, 4) is:

- (a) (3, -10) (b) (2, -10) (c) (-3, 10) (d) (-2, 10)

Ans: (a) (3, -10)

46. The area of a rhombus whose vertices are (3, 0), (4, 5), (-1, 4) and (-2, -1) taken in order, is:

- (a) 12 sq. Units (b) 24 sq. Units (c) 30 sq. Units (d) 32 sq. Units

Ans: (b) 24 sq. Units

47. The distance of the point P(-6, 8) from the origin is

- (a) 8 units (b) $2\sqrt{7}$ units (c) 10 units (d) 6 units

Ans: (c) 10 units

48. The distance between the points (0, 5) and (-5, 0) is

- (a) 5 units (b) $5\sqrt{2}$ units (c) $2\sqrt{5}$ units (d) 10 units

Ans: (b) $5\sqrt{2}$ units

49. The perimeter of a triangle with vertices (0, 4), (0, 0) and (3, 0) is

- (a) 5 (b) 12 (c) 11 (d) $7 + \sqrt{5}$

Ans: (b) 12

$AB = \sqrt{[(0)^2 + (4)^2]} = \sqrt{16} = 4$
 $BC = \sqrt{[(3)^2 + (0)^2]} = \sqrt{9} = 3$
 $CA = \sqrt{[(3)^2 + (4)^2]} = \sqrt{(9 + 16)} = \sqrt{25} = 5$
 Perimeter of the triangle = $AB + BC + CA = 4 + 3 + 5 = 12$

50. The area of a triangle with vertices (a, b + c), (b, c + a) and (c, a + b) is

- (a) $(a + b + c)^2$ (b) 0 (c) $a + b + c$ (d) ab

Ans: (b) 0

51. The point which lies on the perpendicular bisector of the line segment joining the points A(-2, -5) and B(2, 5) is

- (a) (0, 0) (b) (0, 2) (c) (2, 0) (d) (-2, 0)

Ans: (a) (0, 0)

52. If the points A(1, 2), O(0, 0) and C(a, b) are collinear, then

- (a) $a = b$ (b) $a = 2b$ (c) $2a = b$ (d) $a = -b$

Ans: (c) $2a = b$

53. If the points A(6, 1), B(8, 2), C(9, 4) and D(p, 3) are the vertices of a parallelogram, taken in order, then the value of p is

- (a) 4 (b) -6 (c) 7 (d) -2

Ans: (c) 7

54. A line intersects the y-axis and x-axis at the points P and Q, respectively. If (2, -5) is the midpoint of PQ, then the coordinates of P and Q are, respectively

- (a) (0, -5) and (2, 0) (b) (0, 10) and (-4, 0)
(c) (0, 4) and (-10, 0) (d) (0, -10) and (4, 0)

Ans: (d) (0, -10) and (4, 0)

55. The fourth vertex D of a parallelogram ABCD whose three vertices are A(-2, 3), B(6, 7) and C(8, 3) is

- (a) (0, 1) (b) (0, -1) (c) (-1, 0) (d) (1, 0)

Ans: (b) (0, -1)

The coordinates of the midpoint of AC = coordinates of the midpoint of BD

$$[(-2 + 8) / 2, (3 + 3) / 2] = [(6 + x) / 2, (7 + y) / 2]$$

$$6 + x = 6, 7 + y = 6 \quad (x = 0, y = -1)$$

Therefore, the coordinates of the fourth vertex D = (0, -1)

56. If the points (k, 2k), (3k, 3k) and (3, 1) are collinear then what is the value of k.

- (a) -1 / 3 (b) 1 / 3 (c) -3 (d) 3

Ans: (a) -1 / 3

57. Slope of line parallel to y-axis is _____

- (a) Not determined (b) 1 (c) 0 (d) 2

Ans: (a) Not determined

58. Slope of x-axis is _____

- (a) 1 (b) 2 (c) 0 (d) Not determined

Ans: (c) 0

59. The slope of line passing through A (2, 3), B (4, 7) is _____.

- (a) 1 (b) 5 / 2 (c) -4 (d) 2

Ans: (d) 2

60. If slope of line joining the point (2, k) and (-3, 0) is 2 / 5. Then what is the value of k.

- (a) 2 (b) -2 (c) -5 (d) 5

Ans: (a) 2

61. What are the values of m and n if D (m, -2) is the midpoint of the segment joining (-3, n) and (2, -5).

- (a) m = 1, n = -1 / 2 (b) m = -1, n = 1 / 2
(c) m = -1 / 2, n = 1 (d) m = -1 / 2, n = -1

Ans: (c) m = -1 / 2, n = 1

62. The point of concurrence centroid divides the median in the ratio is _____.
(a) 1:2 (b) 2:1 (c) 2:3 (d) 4:2

Ans: (b) 2: 1

63. Angle 6 made by the line with positive direction of X-axis is 45° then the slope of line is _____.
(a) 1 (b) 2 (c) $1/2$ (d) $1/\sqrt{2}$

Ans: (a) 1

64. P divides segment QR internally in the ratio 2:1. $Q = (-5, 8)$, $R = (4, -4)$ then P = _____.
(a) (0, 1) (b) (0, -1) (c) (1, 0) (d) (-1, 0)

Ans: (c) (1, 0)

65. (3, 3), (-4, 1) and (3, -5) are the vertices of a / an _____.

- (a) Right angle triangle (b) Isosceles triangle
(c) Equilateral triangle (d) Scalene triangle

Ans: (b) Isosceles triangle

66. The co-ordinate of the mid-point of the line segment joining the points P (4, -6) and Q (-2, 4) is _____.

- (a) (-1, 1) (b) (1, -1) (c) (3, 5) (d) (-3, -5)

Ans: (b) (1, -1)

67. The ratio in which the point (5, 4) divides the line joining points (2, 1) and (7, 6) is _____.

- (a) 3: 2 (b) 2: 3 (c) 1: 3 (d) 2: 3

Ans: (a) 3:2

68. A (4, 8), B (5, 5), C (2, 4) and D (1, 7) are the vertices of a parallelogram then the co-ordinates of the point of intersection of the diagonals of the parallelogram ABCD is _____.

- (a) (6, 12) (b) (12, 6) (c) (3, 6) (d) (6, 3)

Ans: (c) (3, 6)

69. Distance of point (-3, 4) from the origin is _____.

- (a) 7 (b) 1 (c) 5 (d) -5

Ans: (c) 5

70. Out of the following point _____ lies to the right of the origin on X-axis.

- (a) (-2, 0) (b) (0, 2) (c) (2, 3) (d) (2, 0)

Ans: (d) (2, 0)

71. Segment AB is parallel to Y-axis and co-ordinates of Point A are (1, 3) then co-ordinates of point B can be _____.

- (a) (3, 1) (b) (5, 3) (c) (3, 0) (d) (1, -3)

Ans: (d) (1, -3)

72. P (1, 2) is the midpoint of segment AB. If B (4, 3), what are the co-ordinates of the point A?

- (a) (6,1) (b) (1,6) (c) (-1,6) (d) (-6,1)

Ans: (d) (-6, 1)

73. What is the distance between the points A (4, 5) and B (2, 5)?

- (a) 4 (b) 2 (c) 16 (d) (1)

Ans: (b) 2

74. What will be the value of y, if the ratio in which the point (34, y) divides the line segment joining the points A (-1, 4) and B (6, 5) is 1:3?

- (a) $y = 9 / 2$ (b) $y = 5 / 2$ (c) $y = 9 / 4$ (d) $y = 5 / 2$

Ans: (a) $y = 9 / 2$

$$x = \frac{mx_2 + nx_1}{m+n} \text{ and } y = \frac{my_2 + ny_1}{m+n}$$

$$x = \frac{1(6) + 3(-1)}{1+3} = \frac{3}{4} \quad , \quad y = \frac{1(6) + 3(4)}{1+3} = \frac{18}{4} = \frac{9}{2}$$

75. What will be the coordinates of the midpoint of the line segment joining the points (-5, 10) and (15, 2)?

- (a) (-5, -6) (b) (-5, 6) (c) (5, 6) (d) (5, -6)

Ans: (c) (5, 6)

Section Formula is $x = \frac{mx_2 + nx_1}{m+n}$ and $y = \frac{my_2 + ny_1}{m+n}$

$$x = \frac{1(-5) + 1(15)}{2} = \frac{10}{2} = 5 \quad , \quad y = \frac{1(2) + 1(10)}{2} = \frac{12}{2} = 6 \quad \text{Hence, the point is (5,6).}$$

76. What will be the coordinates of the point which divides the line segment joining the points A (-2, 2) and B (-1, 5) in the ratio 2: 5?

- (a) $(-4 / 3, -20 / 9)$ (b) $(-4 / 3, 20 / 9)$
(c) $(4 / 3, 20 / 9)$ (d) $(4 / 3, -20 / 9)$

Ans: (b) $(-4 / 3, 20 / 9)$

Using, section formula $x = \frac{mx_2 + nx_1}{m+n}$ and $y = \frac{my_2 + ny_1}{m+n}$

$$x = \frac{2(-1) + 5(-2)}{2+5} = \frac{-12}{7} = \frac{-4}{3} \quad , \quad y = \frac{2(5) + 5(2)}{2+5} = \frac{20}{7}$$

Hence, the point is $(\frac{-4}{3}, \frac{20}{7})$.

77. The two vertices of $\triangle ABC$ are given by A (-3, 0) and B (-8, 5) and its centroid is (-2, 1). What will be the coordinates of the third vertex C?

- (a) (-5, -2) (b) (5, 2) (c) (-5, 2) (d) (5, -2)

Ans: (d) (5, -2)

78. The coordinates of the ends of the diameter AB of a circle are A (-4, 7) and B (4, 7). What will be the coordinates of the centre of the circle?
(a) (0, -8) (b) (0, 8) (c) (0, 7) (d) (0, -7)

Ans: (c) (0, 7)

79. The coordinates of one end of the diameter AB of a circle are A (-2, -3) and the coordinates of diameter are (-2, 0). What will be the coordinates of B?
(a) (2, -3) (b) (-2, 3) (c) (2, 3) (d) (-2, -3)

Ans: (b) (-2, 3)

80. What will be the centroid of the $\triangle ABC$ whose vertices are A (-2, 4), B (0, 0) and C (4, 2)?
(a) $(2/3, 2)$ (b) $(2/3, 1)$ (c) $(2/5, 2)$ (d) $(1/3, 2)$

Ans: (a) $(2/3, 2)$

81. What will be the coordinates of B, if the point C $(29/7, 46/7)$, divides the line segment joining A (5, 8) and B (a, b) in the ratio 2:5?
(a) $a = 2, b = 3$ (b) $a = -2, b = 3$ (c) $a = 2, b = -3$ (d) $a = -2, b = -3$

Ans: (a) $a = 2, b = 3$

82. What will be the area of a triangle whose vertices are (3, 1), (0, 4) and (5, 9)?
(a) 5 units (b) -5 units (c) -15 units (d) 15 units

Ans: (c) -15 units

83. What will be the area of the quadrilateral ABCD whose vertices are A (8, 6), B (9, 0), C (1, 2) and D (3, 4)?
(a) -20 units (b) 26 units (c) 23 units (d) 3 units

Ans: (b) 26 units

84. What will be the area of triangle PQR formed by joining the midpoints of the sides of the triangle whose vertices are (3, 3), (9, 9) and (4, 6)?
(a) 12 units (b) 32 units (c) 35 units (d) 52 units

Ans: (b) 32 units

85. If the points A, B, C is collinear then the area of the triangle will be zero.
(a) True (b) False (c) Infinity (d) None of these

Ans: (a) True

86. What will be the area of triangle formed by joining the points P (2, 3), Q (4, 6) and R (6, 9)?
(a) 0 (b) 2 (c) 3 (d) 4

Ans: (a) 0

87. What will be the value of p, for which the points (-3, p), (-1, 2) and (1, 1) are collinear?
(a) $p = -4$ (b) $p = 4$ (c) $p = 3$ (d) $p = -3$

Ans: (c) $p = 3$

88. The points A (3, 0), B (4, 5) and C (6, 7) are collinear.

- (a) True (b) False (c) Non – Collinear (d) None

Ans: (b) False

89. Three points A (x_1, y_1), B (x_2, y_2) and C (x_3, y_3) are collinear only when $\frac{1}{2} \{x_1 (y_2 - y_3) + x_2 (y_3 - y_1) + x_3 (y_1 - y_2)\} = 0$.

- (a) False (b) True (c) Can't Determine (d) None

Ans: (b) True

90. If A (x_1, y_1), B (x_2, y_2) and C (x_3, y_3) are the vertices of a triangle ABC, then its area is given by $\frac{1}{2} \{x_1 (y_2 - y_3) + x_2 (y_3 - y_1) + x_3 (y_1 - y_2)\}$

- (a) True (b) False (c) Can't Determine (d) None

Ans: (a) True

91. If the points A (a, 0), B (0, b) and C (c, 0) are collinear then $a = -c$.

- (a) False (b) True (c) $a = -b$ (d) None

Ans: (a) False

92. The distance of a point from the Y - axis is called its -----

- (a) Ordinate (b) Abscissa (c) Origin (d) None of these

Ans: (b) Abscissa

93. The distance of a point from the X - axis is called its -----

- (a) Ordinate (b) Abscissa (c) Origin (d) None of these

Ans: (a) Ordinate

94. A linear equation in two variables of the form $ax + by + c = 0$, when represented graphically gives a -----

- (a) Parabola (b) Circle (c) Straight line (d) None of these

Ans: (c) Straight line

95. The graph of a quadratic equation is a -----

- (a) Parabola (b) Circle (c) Straight line (d) None of these

Ans: (a) Parabola

96. The area of the triangle with vertices at the points (a, b + c), (b, c + a), and (c, a + b) is -----

- (a) $(a + b + c)$ Sq. units (b) $(a + b - c)$ Sq. units
(c) $(a - b + c)$ Sq. units (d) 0

Ans: (d) 0

97. The horizontal and vertical lines drawn to determine the position of a point in a Cartesian plane are called -----

- (a) Intersecting lines (b) Traversals
(c) Perpendicular lines (d) X – axis and Y – axis

Ans: (d) X – axis and Y – axis

98. The points (1, 1), (-2, 7) and (3, -3) are

- (a) Vertices of an equilateral triangle (b) Collinear
(c) Vertices of an isosceles triangle (d) None of these

Ans: (b) Collinear

99. The distance of the point P (2, 3) from the X – axis is

- (a) 2 (b) 3 (c) 1 (d) 5

Ans: (b) 3

100. The distance between the point P (1, 4) and Q (4, 0) is

- (a) 4 (b) 5 (c) 6 (d) $3\sqrt{3}$

Ans: (b) 5